

FusionMedAI: Ocular Fundus Dataset Exploratory Summary

EDA Version: 1.0.0 | Fingerprint: df102971518b3651...

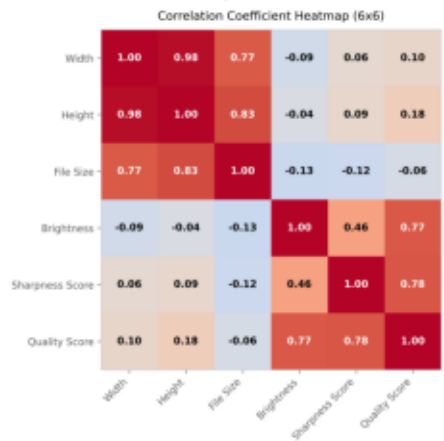
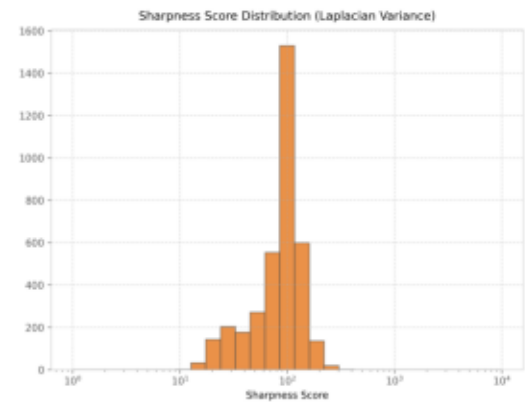
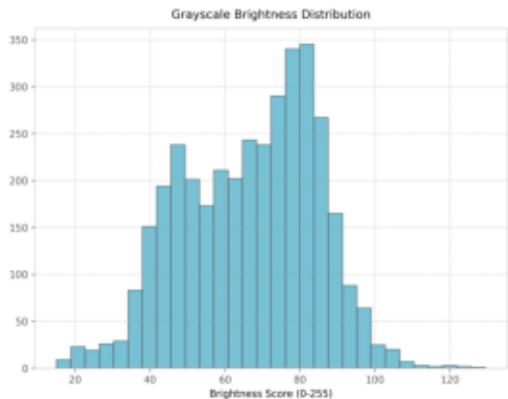
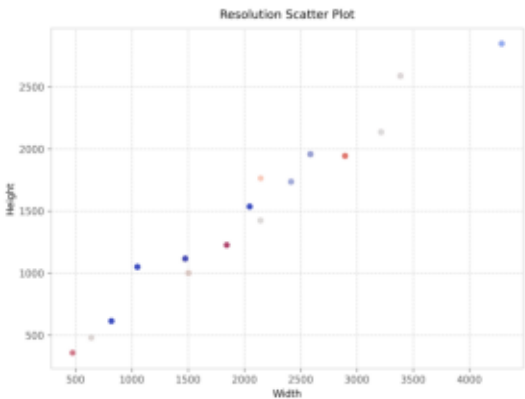
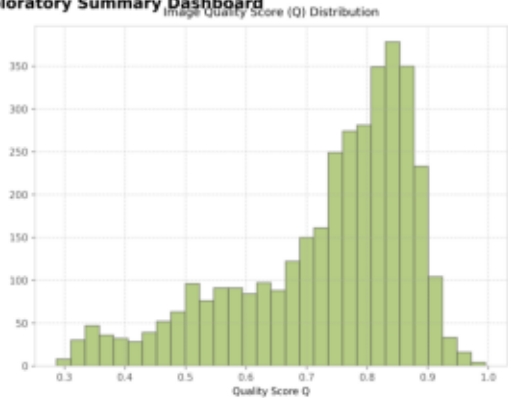
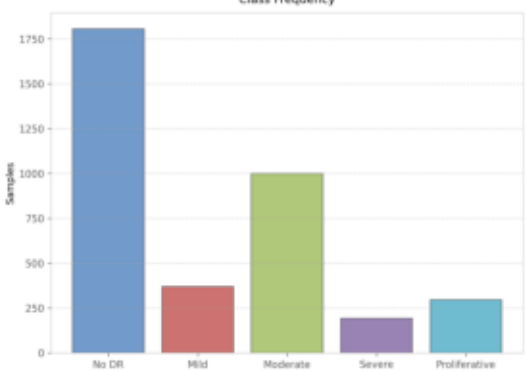
Metric	Value
Total Labeled Images	3662
Ocular Diagnoses Classes	5 Staging Classes
Average Dimensions	2015.2 x 1526.8 pixels
Resolution Extrema	Min: 474x358 Max: 4288x2848
Average File Size	2294.2 KB
Average Aspect Ratio	1.28
Average Quality Score (Q)	0.7332
Global RGB Channel Mean	R: 0.4216 G: 0.2238 B: 0.0722
Global RGB Channel Std	R: 0.2765 G: 0.1501 B: 0.0805
Quality Outlier Flags	Dark threshold < 37.78 Sharpness threshold < 24.37
Corrupted / Missing Files	Corrupted: 0 Missing: 0
Extraction Runtime	77.27 seconds

FusionMedAI: Candidate Preprocessing & Augmentation Parameters

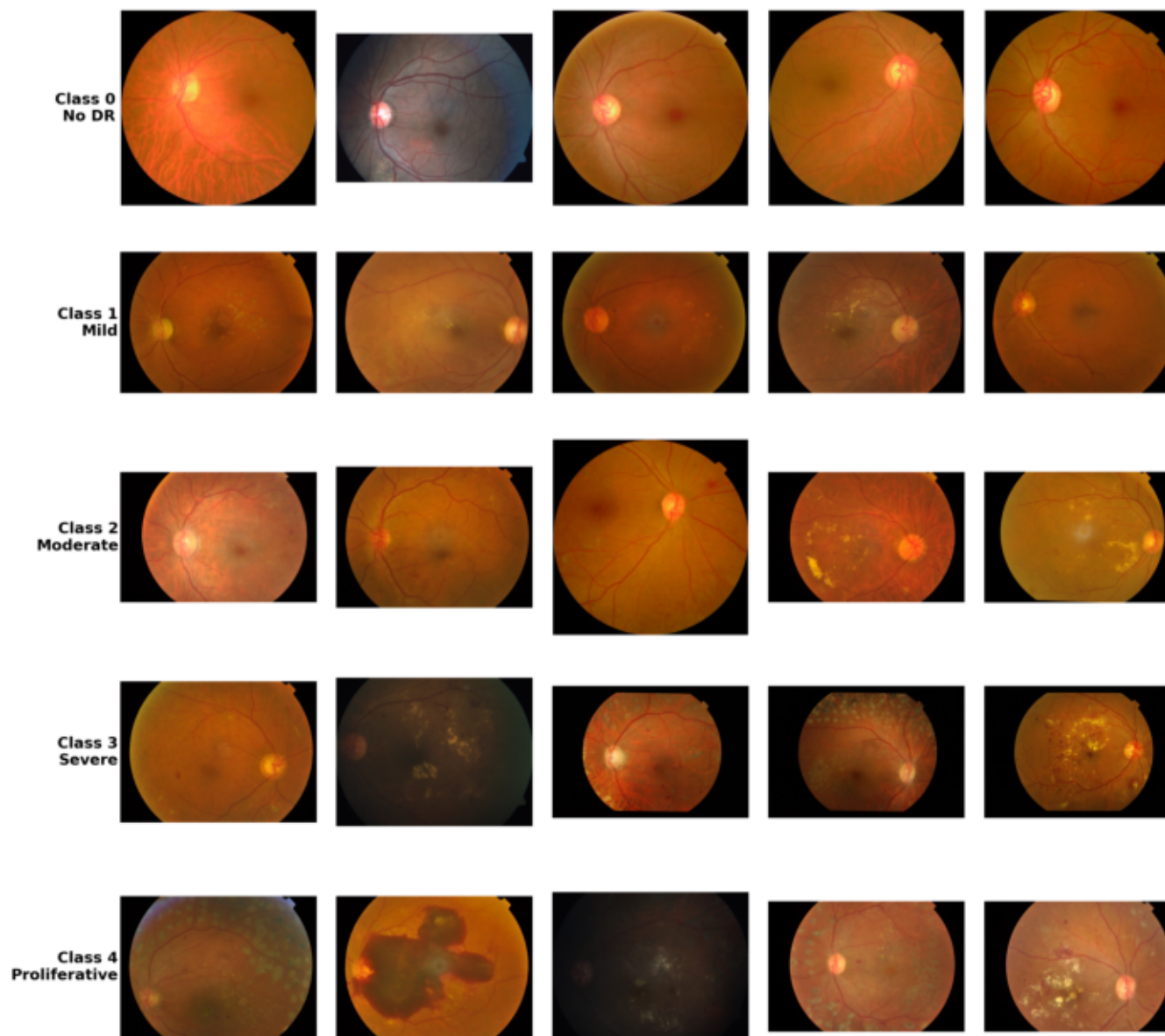
Parameters established to be evaluated experimentally during model training:

Hyperparameter Choice	Candidate Configurations to Benchmark
Candidate Image Sizes	[224, 384, 512] pixels
Candidate Normalizations	[ImageNet, Dataset] Horizontal Flip: [True, False]
Candidate Augmentations	Rotation Degrees: [0, 15, 30] Color Jitter: [True, False] Random Crop: [True, False]
Dark Brightness Threshold	< 37.782003
Bright Brightness Threshold	> 92.419779
Sharpness Threshold (Laplacian)	< 24.368078
File Size Thresholds	Min: 319.00 KB Max: 5525.90 KB

FusionMedAI: Retina Dataset Exploratory Summary Dashboard

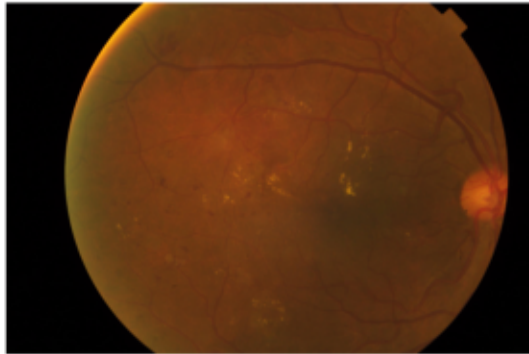


Deterministic Retina Image Sample Grid (5x5)

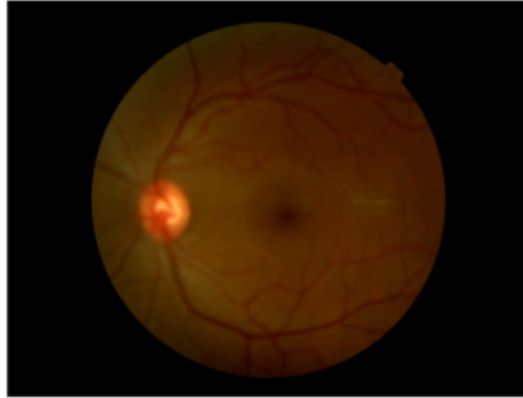


Ocular Fundus Quality Examples and Outlier Cases

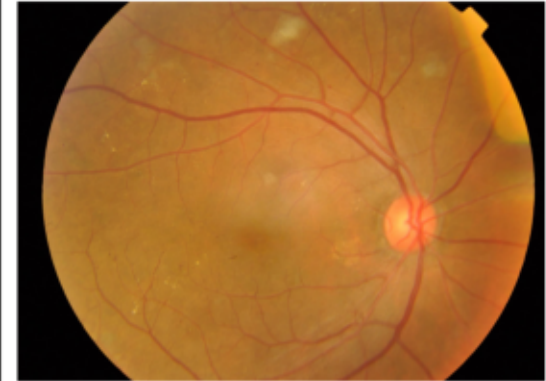
Good Baseline
ID: 000c1434d8d7
Brightness: 51.3 | Sharpness: 111.1
3216x2136 | 3143 KB



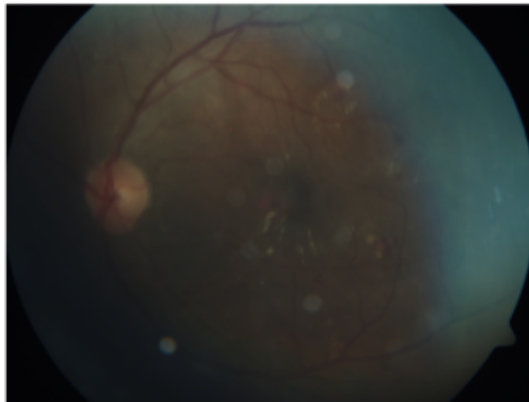
Dark Outlier
ID: 005b95c28852
Brightness: 22.7 | Sharpness: 49.7
2048x1536 | 1777 KB



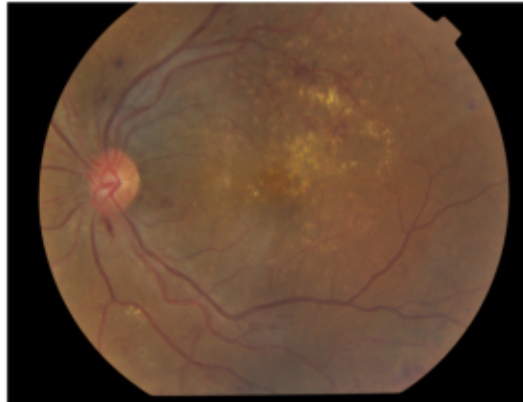
Bright Outlier
ID: 00e4ddff966a
Brightness: 107.5 | Sharpness: 170.8
2416x1736 | 2183 KB



Blurry Outlier
ID: 00b74780d31d
Brightness: 48.8 | Sharpness: 20.5
2588x1958 | 4946 KB



Large Size File
ID: e265c870f9b3
Brightness: 71.4 | Sharpness: 178.4
3388x2588 | 7494 KB



Small Size File
ID: 7efc91af4ae6
Brightness: 67.1 | Sharpness: 78.3
474x358 | 217 KB

